Understanding the Path to Higher Earnings: A Complex Systems Perspective

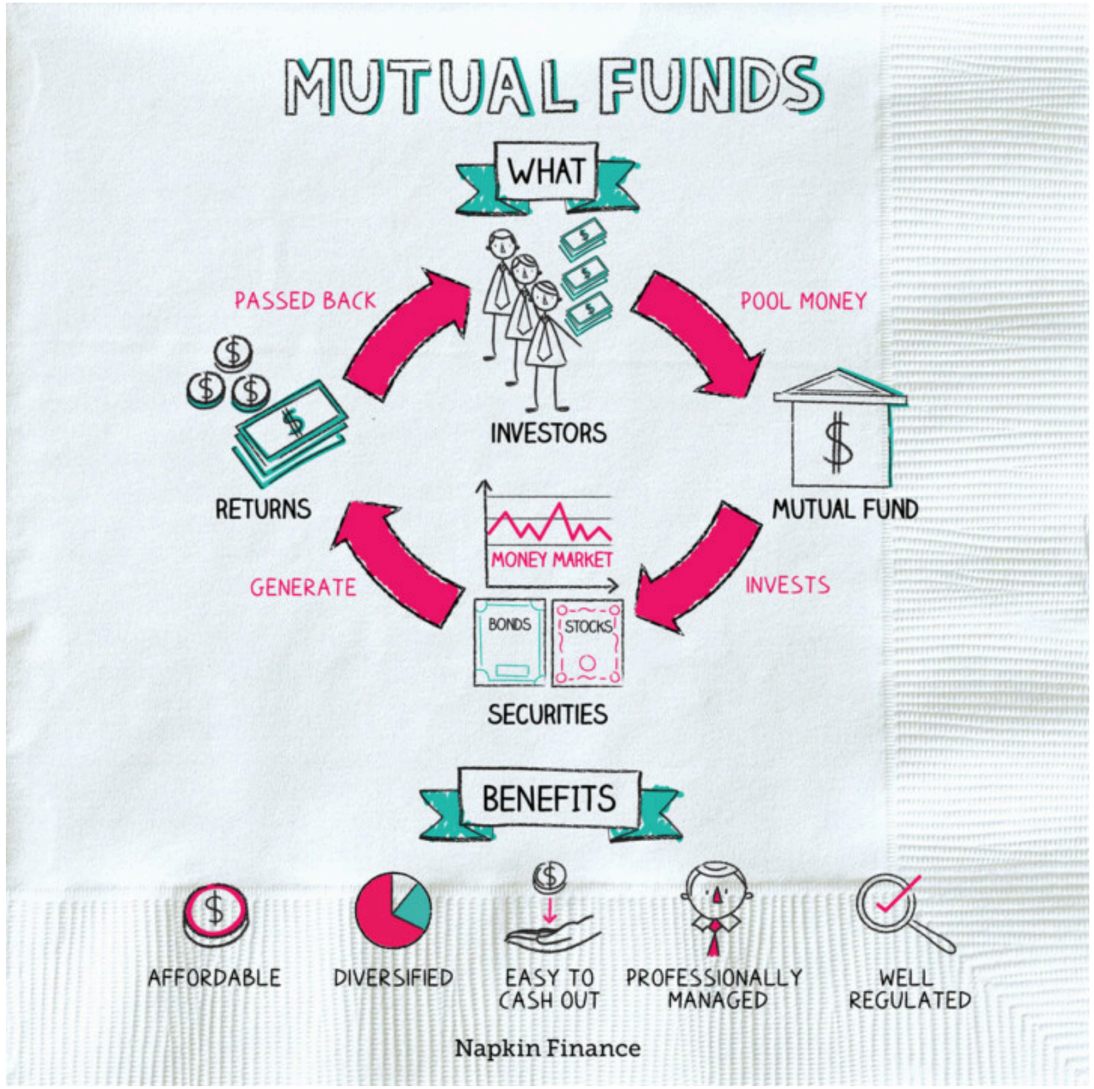
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Mutual Funds

- Mutual funds are pooled investments managed by **professional money managers**.

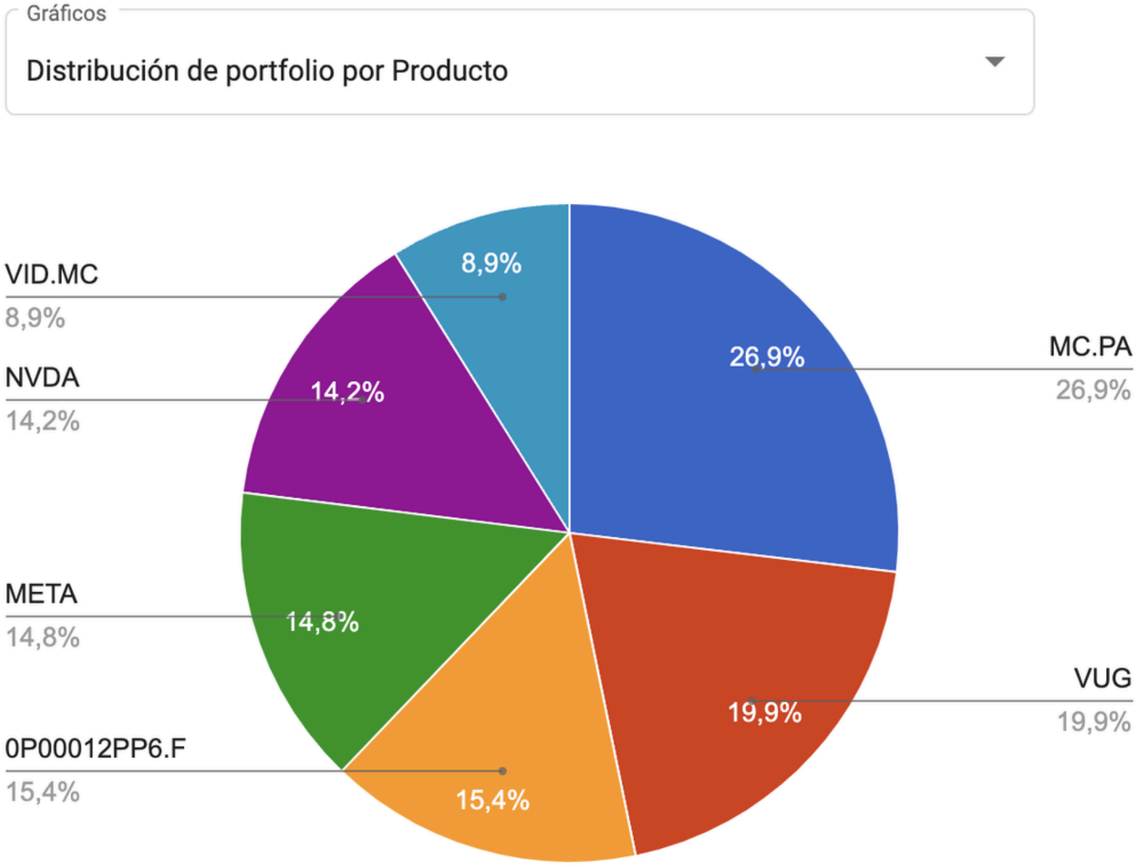


Active Mutual Funds

- **Pick** and **choose** investments, with the aim of delivering a performance that **beats** the fund's stated benchmark or index.

Active Mutual Funds

- These funds hold a **portfolio** of investments across various companies.
- The funds may have a **vocation** to certain sectors



Data

We have data from **spanish** active funds in the period (2000 - 2022).



IBEX 35

Data

We have data from **spanish** active funds in the period (2000 - 2022).



IBEX 35

Constraints:

- 60% investment in Euros.
- No more than 10% in a single company.
- 75% mutual equity funds.

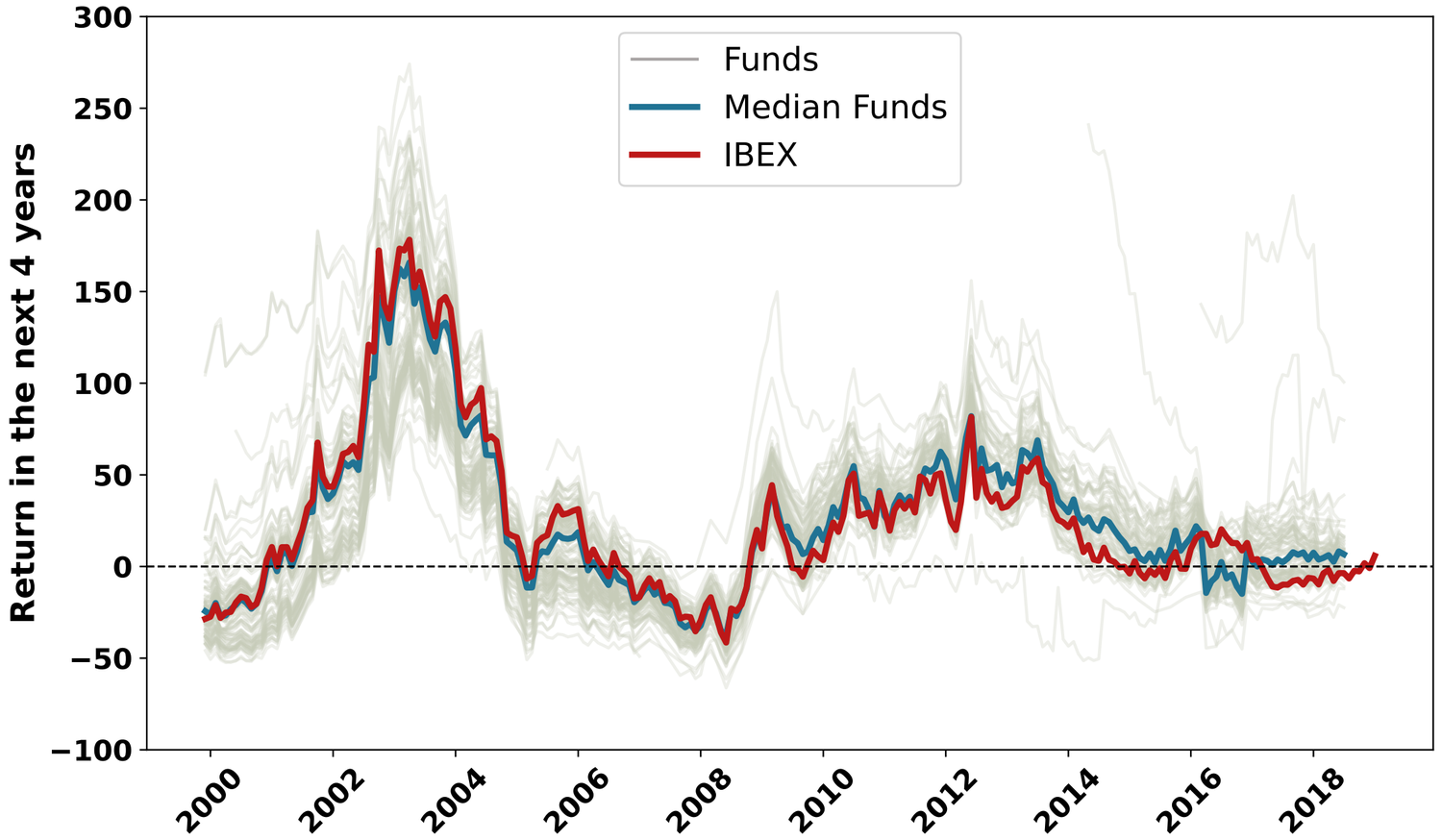
Question

Do spanish active funds overperform IBEX 35?

Funds vs IBEX

In each point of time, we compute the return in the **next 4 years.**

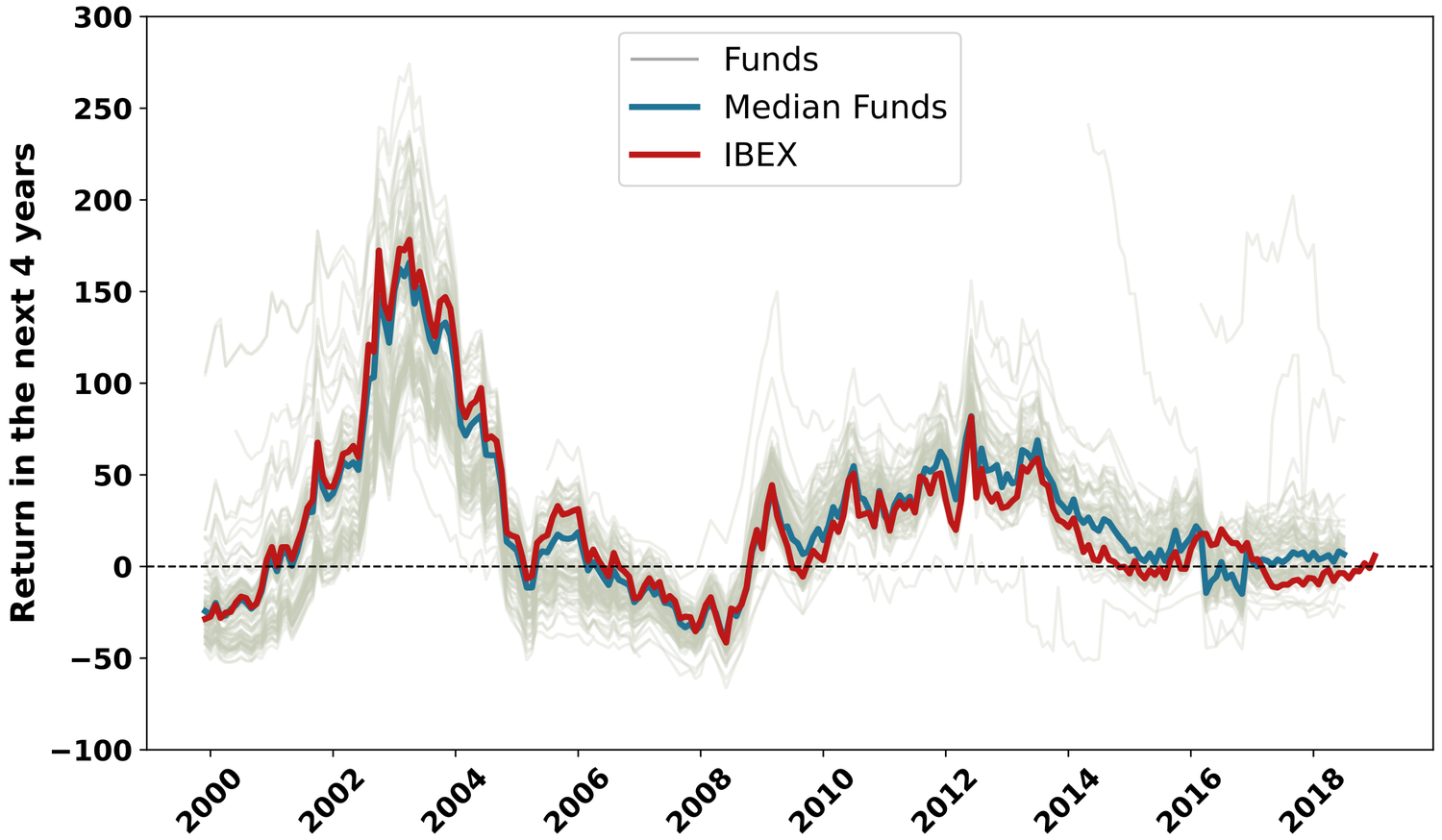
If you invest now, **how much will you make** in 4 years?



Funds vs IBEX

The **median returns** does **not overperform** IBEX-35 before 2010

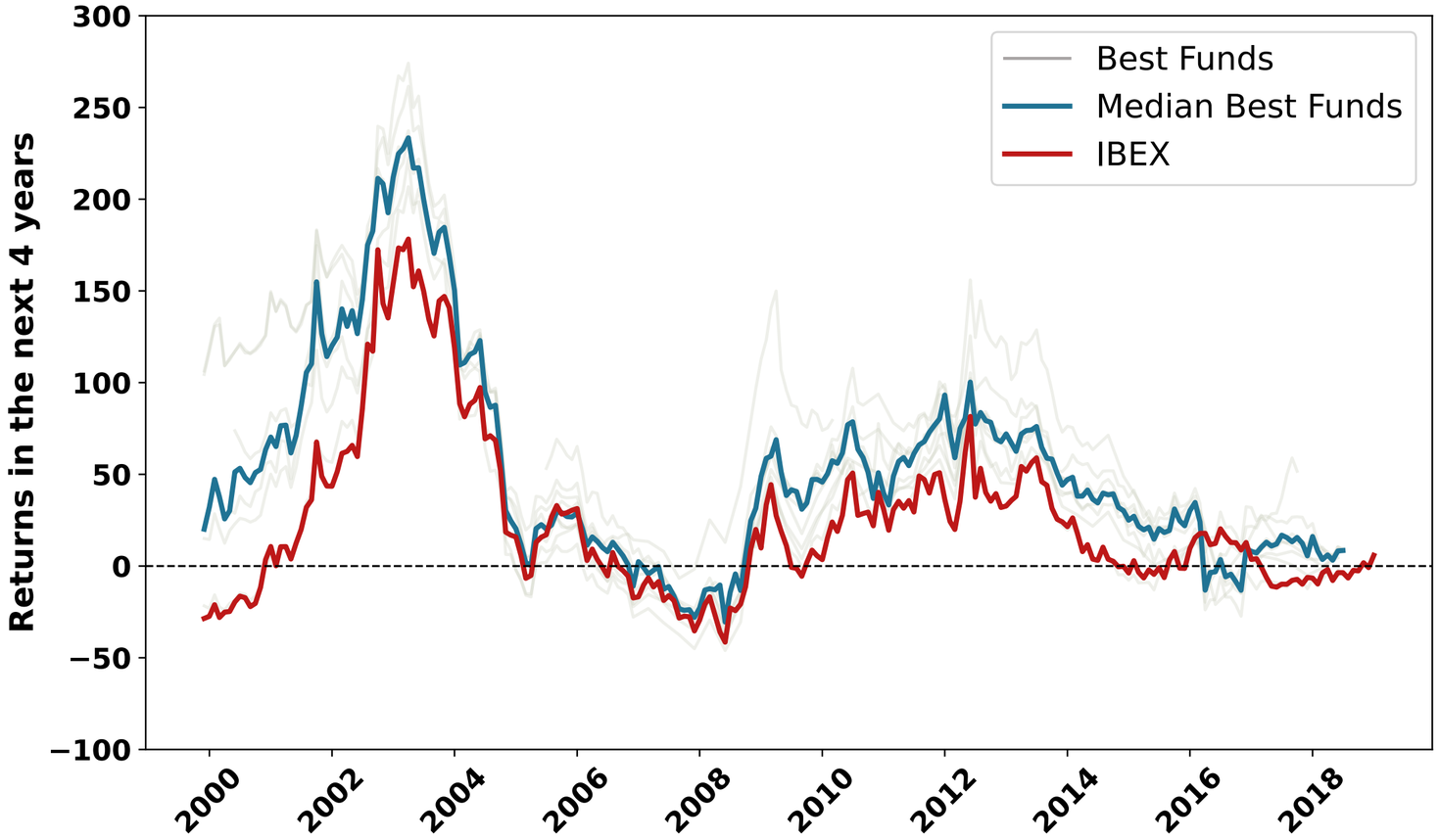
After 2010, the **median** is a little bit **better**.



Funds vs IBEX

We pick the best funds:

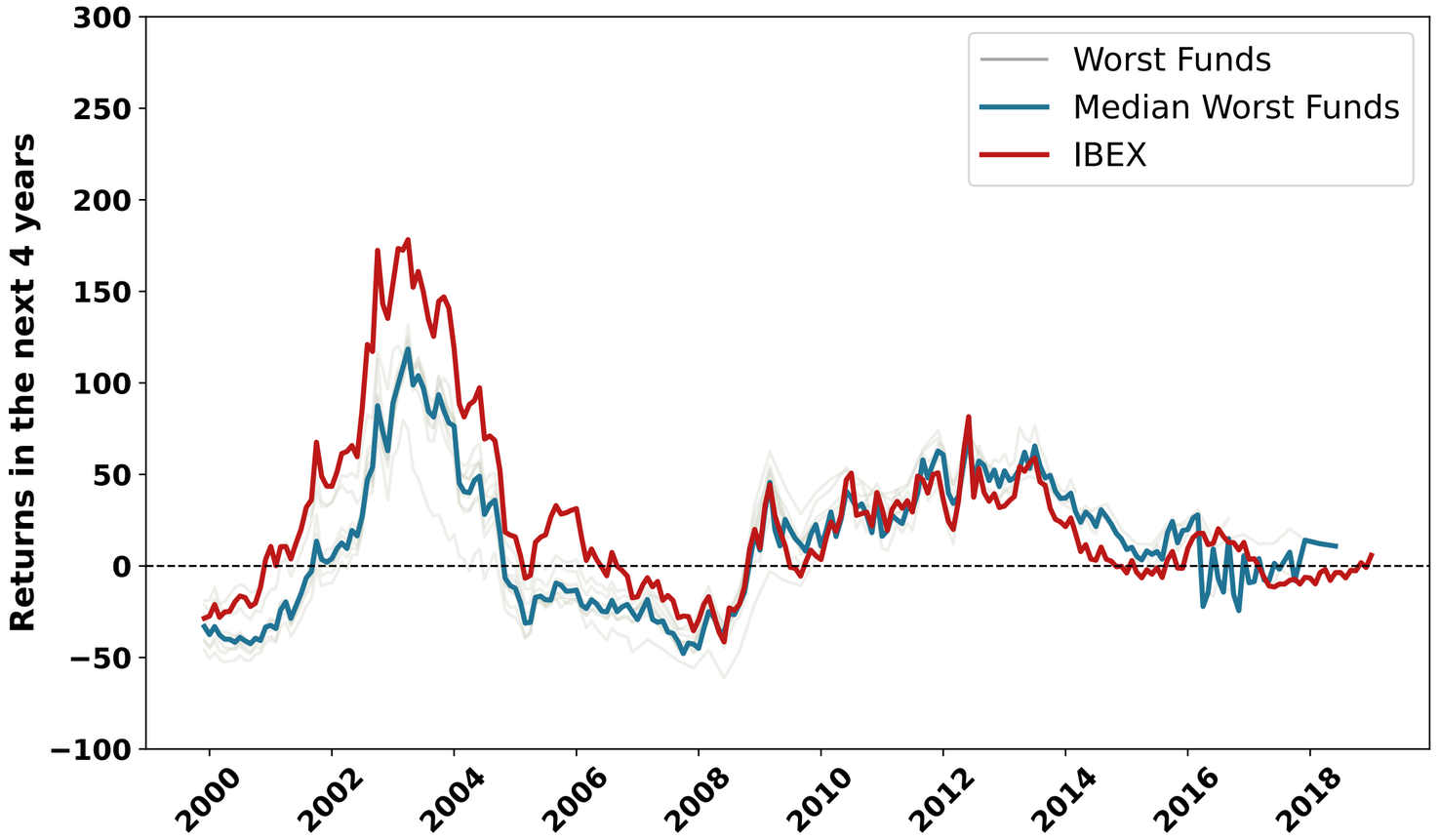
- Over 65 percentile (17.5 years) **lifespan.**
- Over 90 percentile **average return over** IBEX.



Funds vs IBEX

We pick the worst funds:

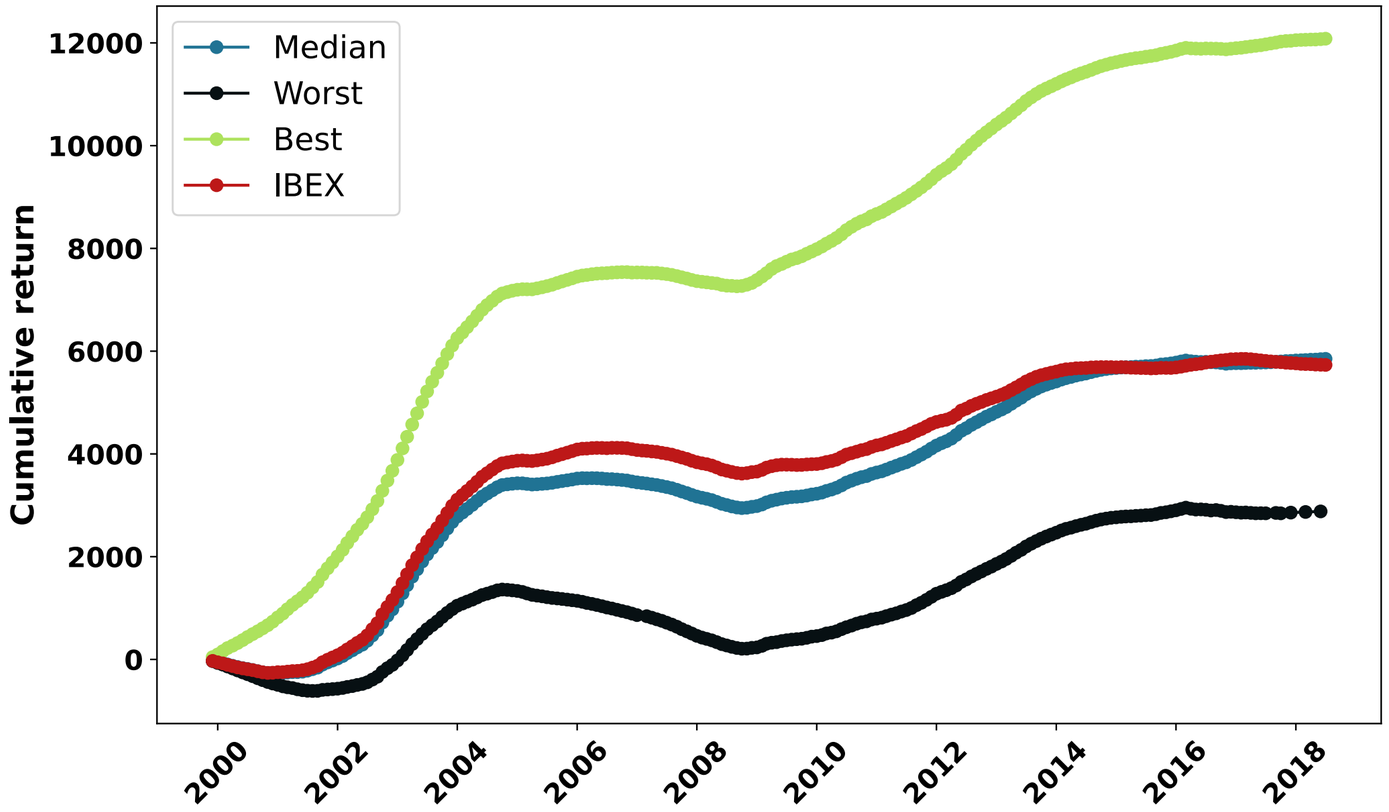
- Over 65 percentile (17.5 years) **lifespan.**
- Under 25 percentile **average return over** IBEX.



Funds vs IBEX

There are periods when **all** funds perform well, but the best ones **outperform** the rest.

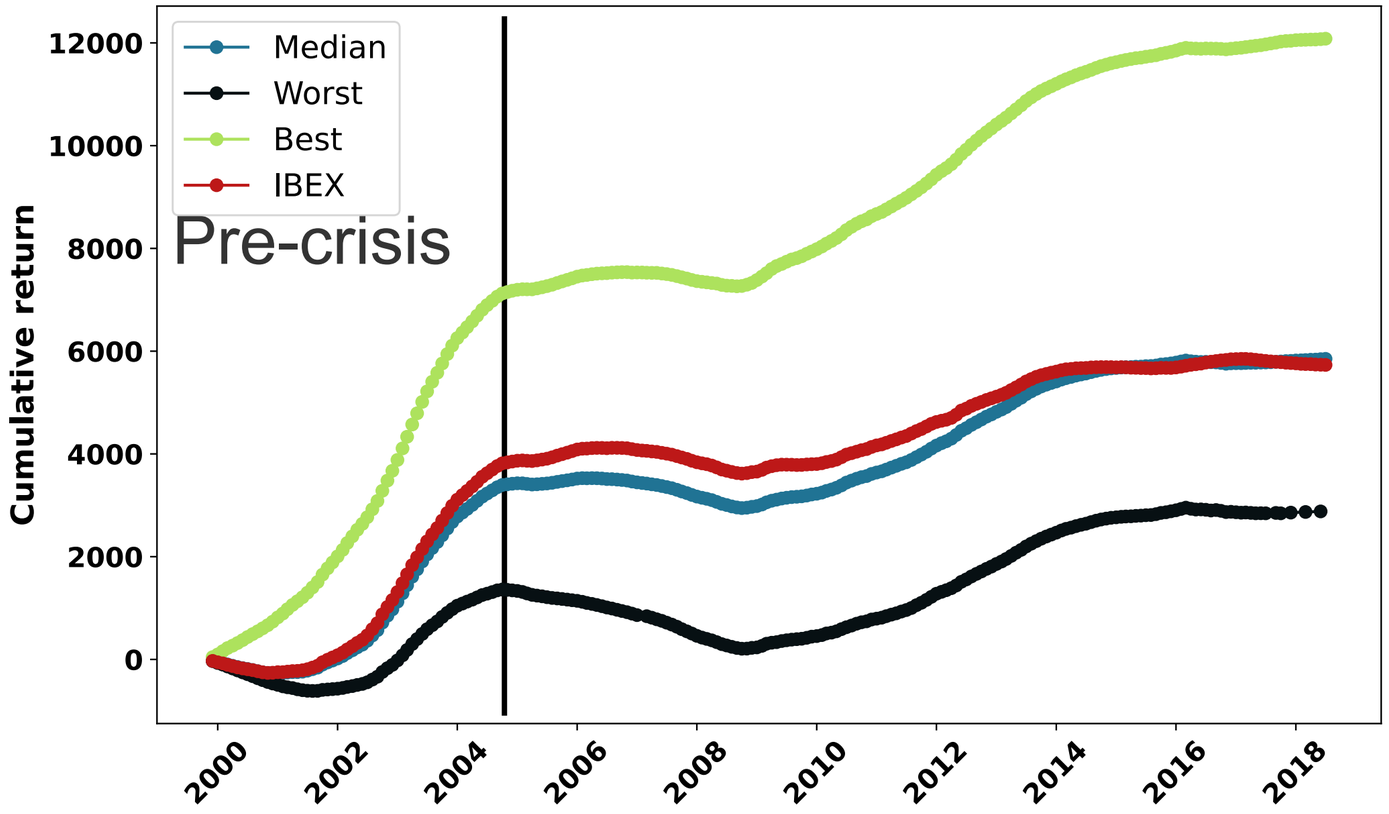
In crisis, the key is **not to lose.**



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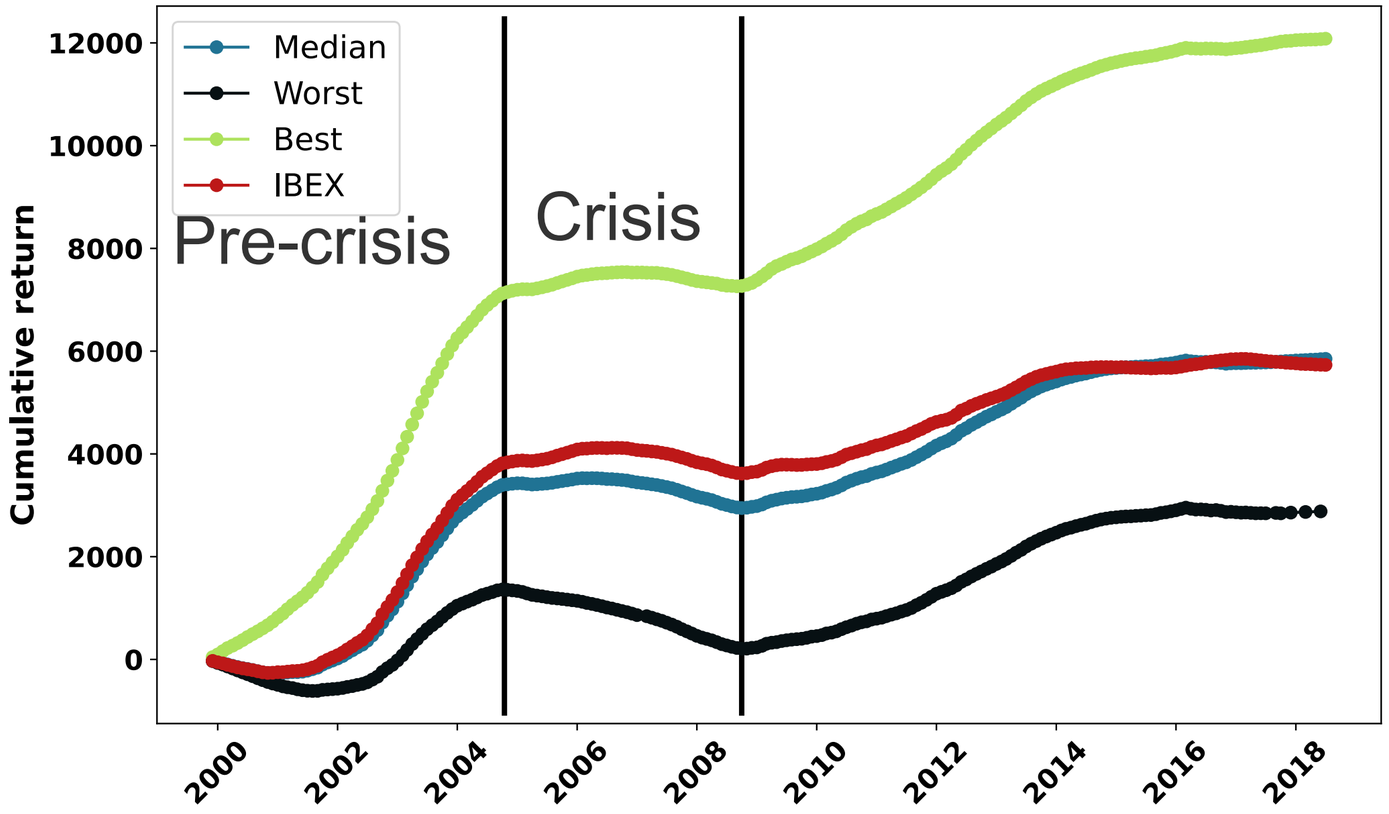
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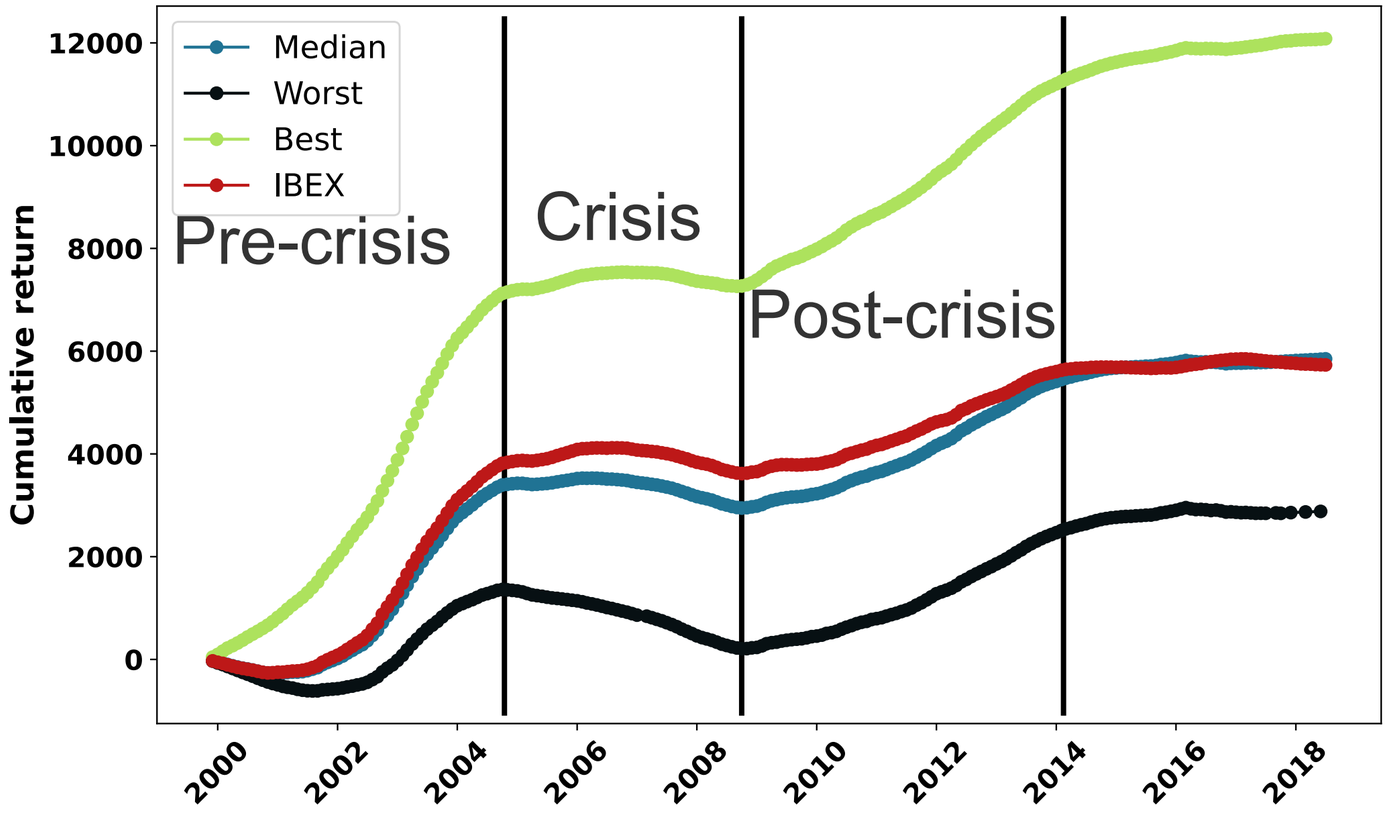
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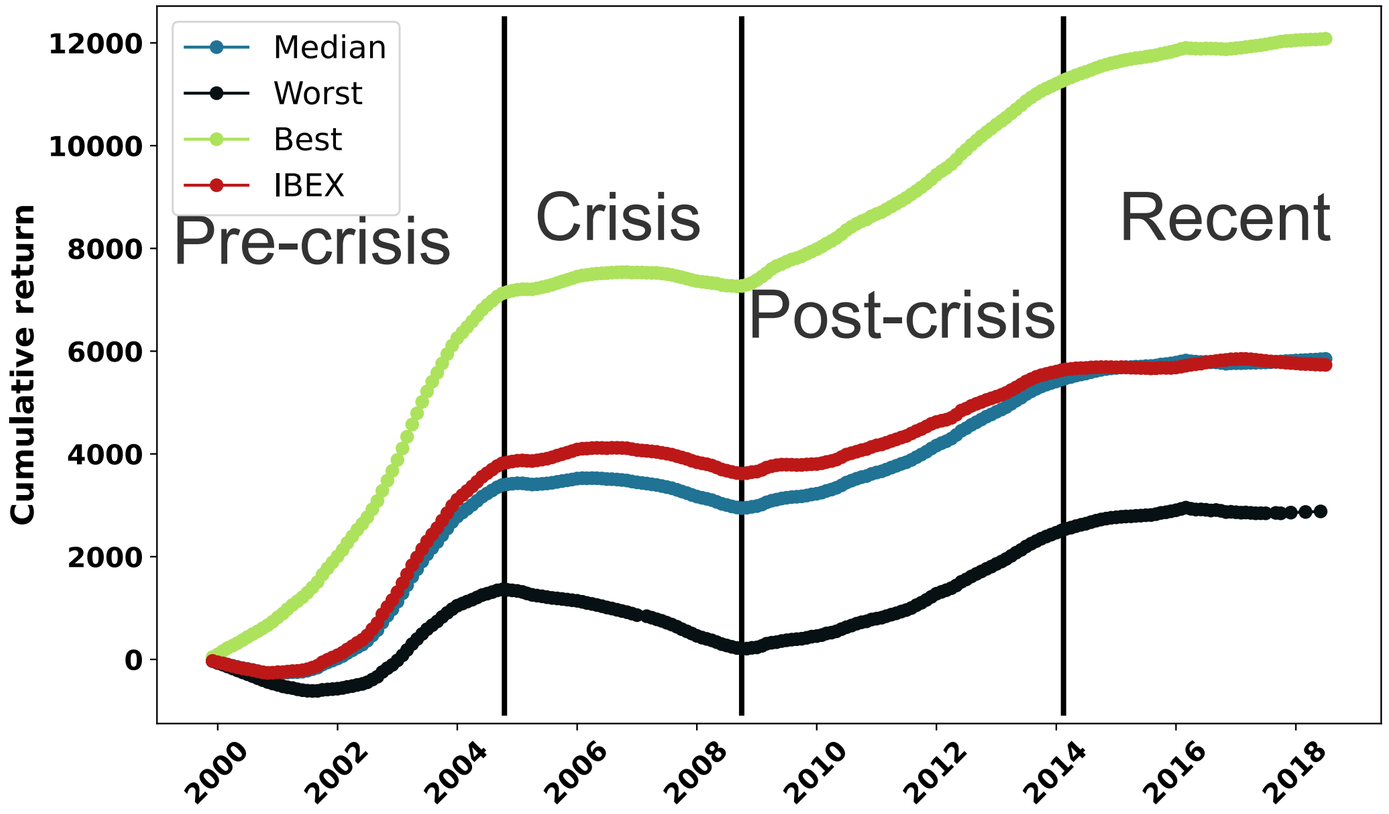
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Funds vs IBEX

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Fund analysis

What is the **composition difference** between those funds and IBEX?

Fund analysis

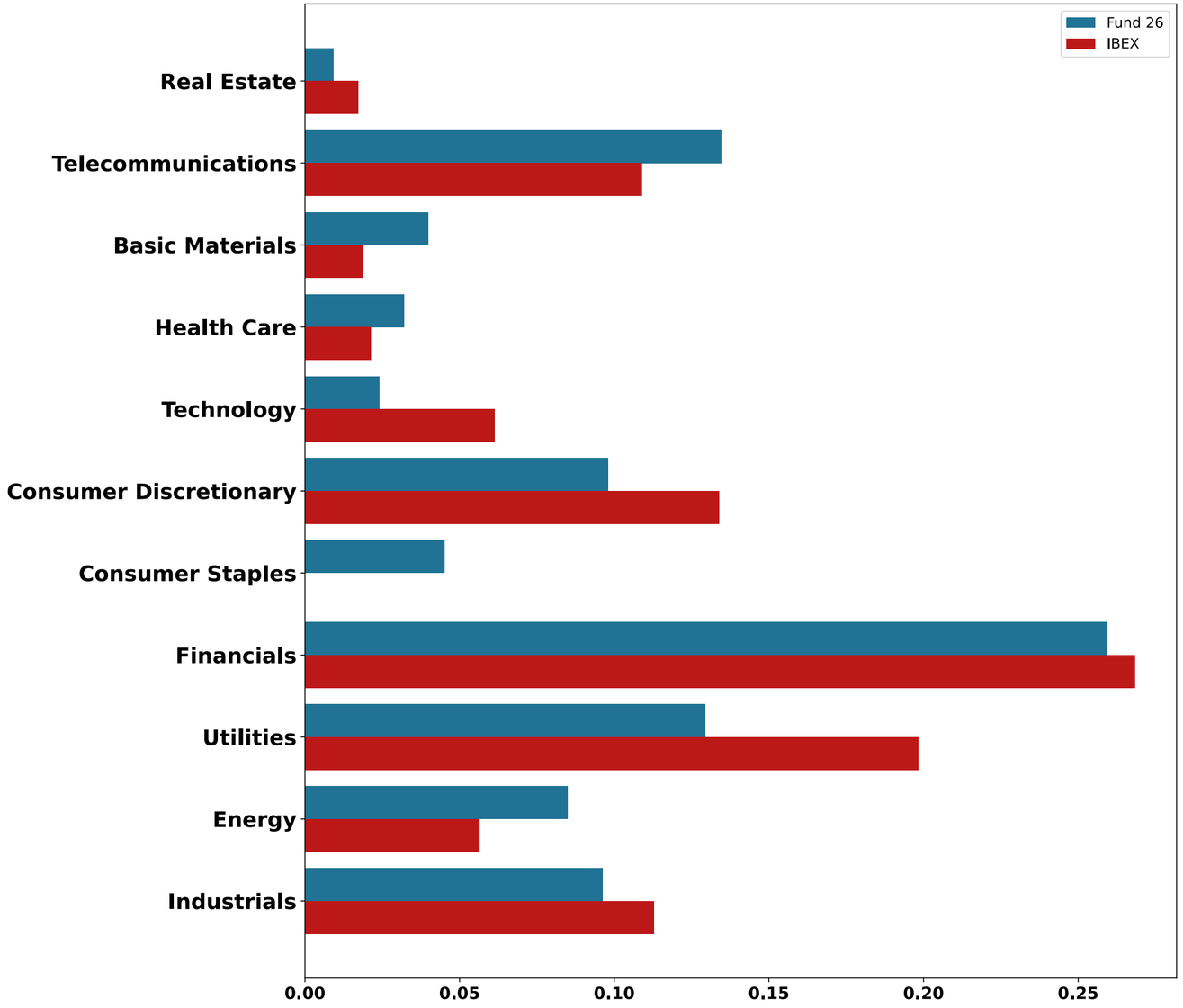
What is the **composition difference** between those funds and IBEX?

We can compare their **distribution by sector**.

Fund analysis

They split their investments into **11 different sectors.**

We can compute the **difference** between the funds and IBEX.



Fund analysis

They split their investments into **11 different sectors.**

We can compute the **difference** between the funds and IBEX.

We use **Jensen Shanon Divergence**

$$JSD(P||Q) = \frac{1}{2}D_{KL}(P||M) + \frac{1}{2}D_{KL}(Q||M)$$

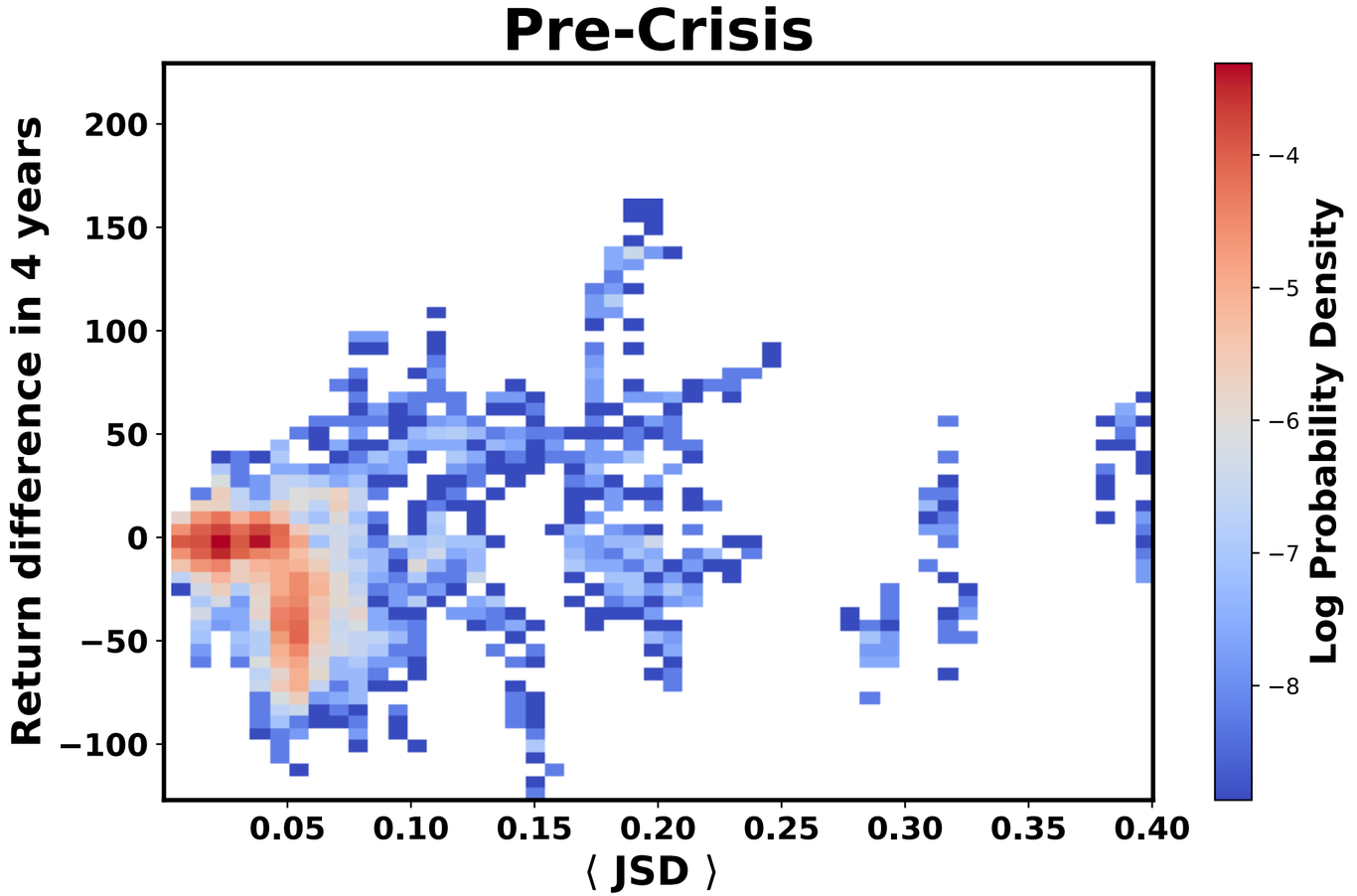
Where

$$D_{KL}(P||Q) = \sum_{x \in X} P(x) log\left(\frac{P(x)}{Q(x)}\right)$$

Fund analysis

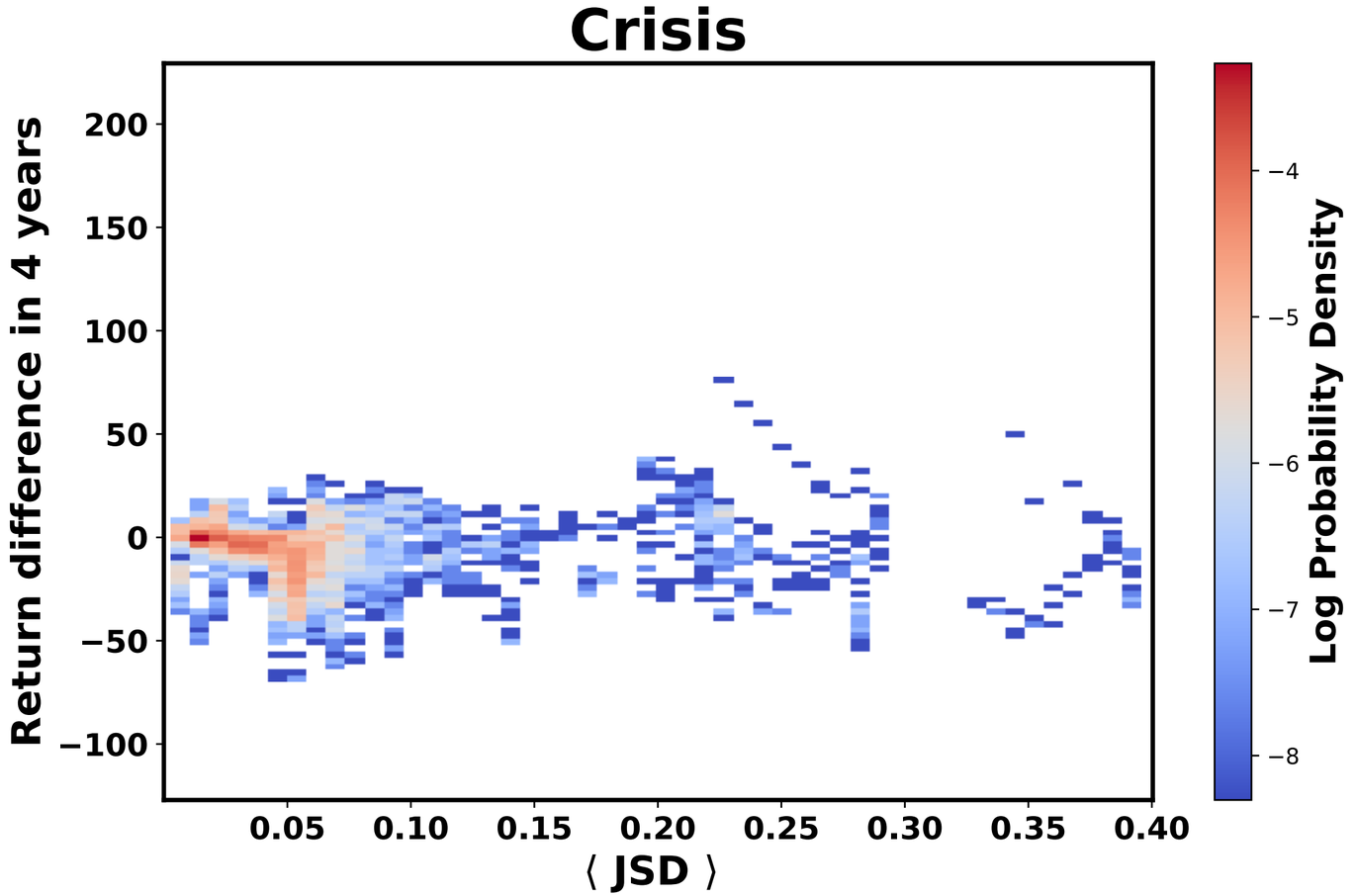
In order to **win** more you should **differ** from IBEX.

There is a **hotspot** over 0.05 of funds not so far from IBEX but they **perform worse.**



Fund analysis

The margins to **win/lose** have **narrowed**

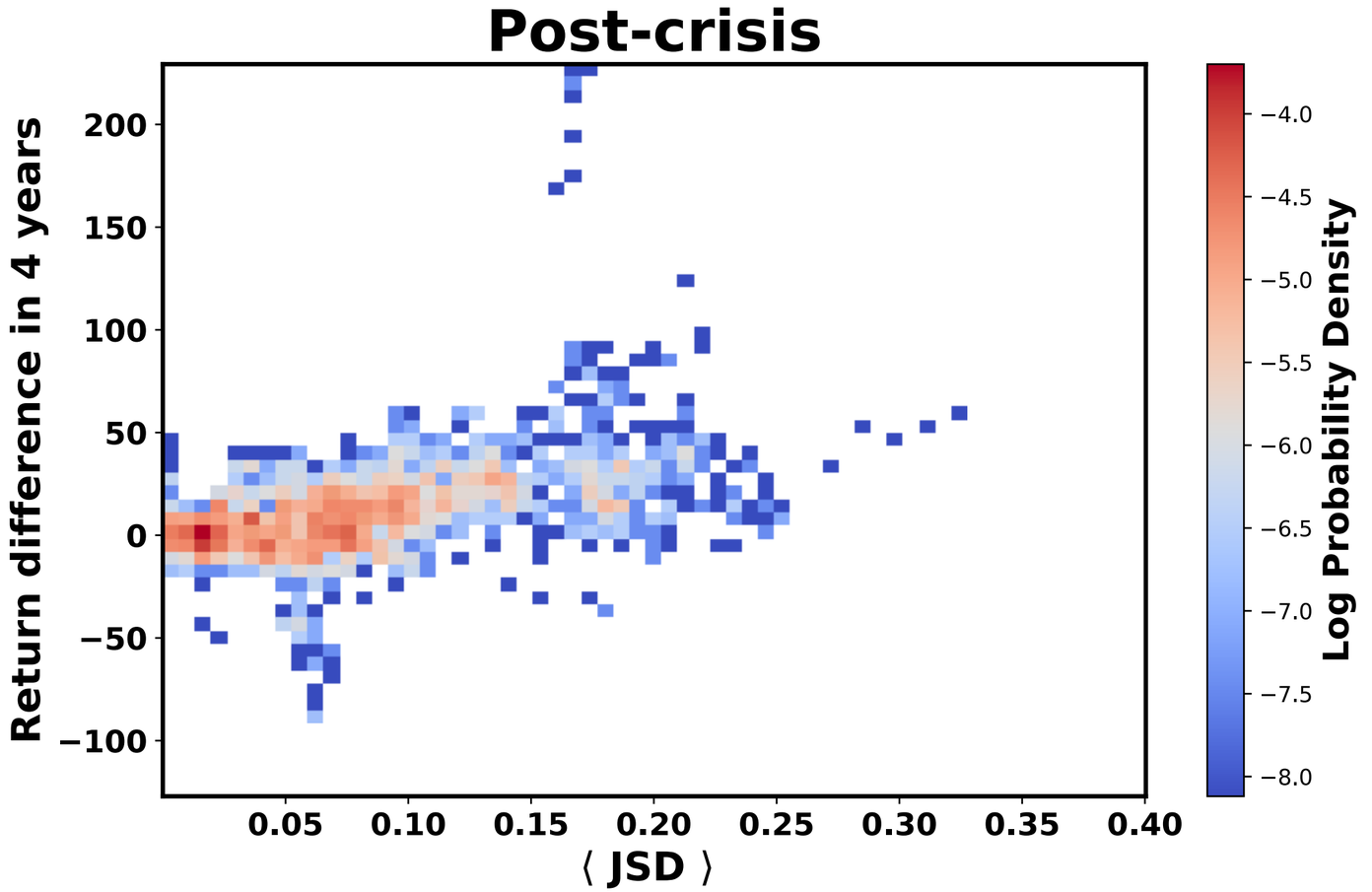


Fund analysis

There is **not** a hotspot in the **underperform** area.

There is a **positive correlation** between **distance and return** over IBEX.

There are **not** funds that **differ a lot.**

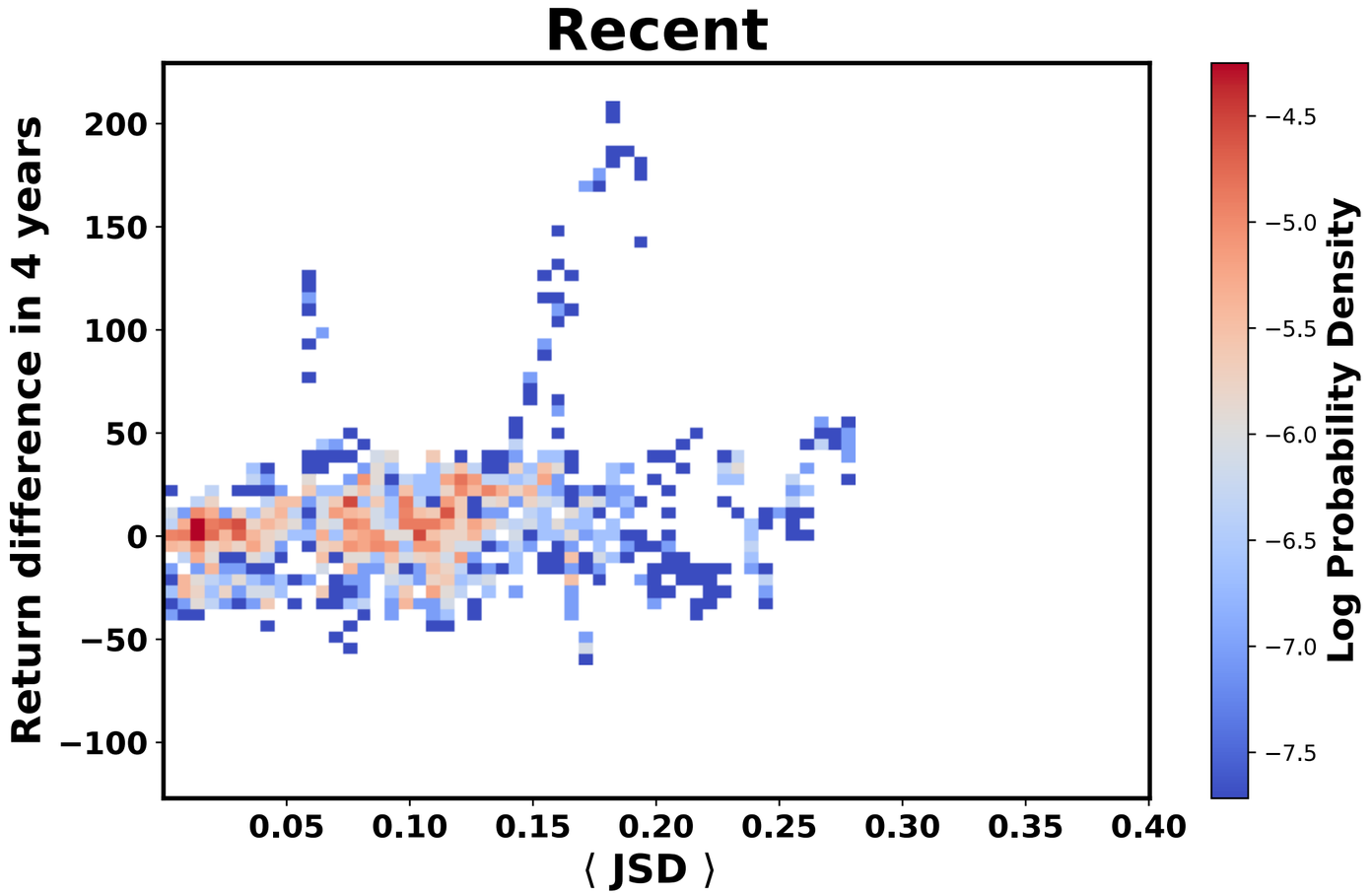


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Null model

What if we **pick** the investments totally **random**?

What if we **create random funds**?

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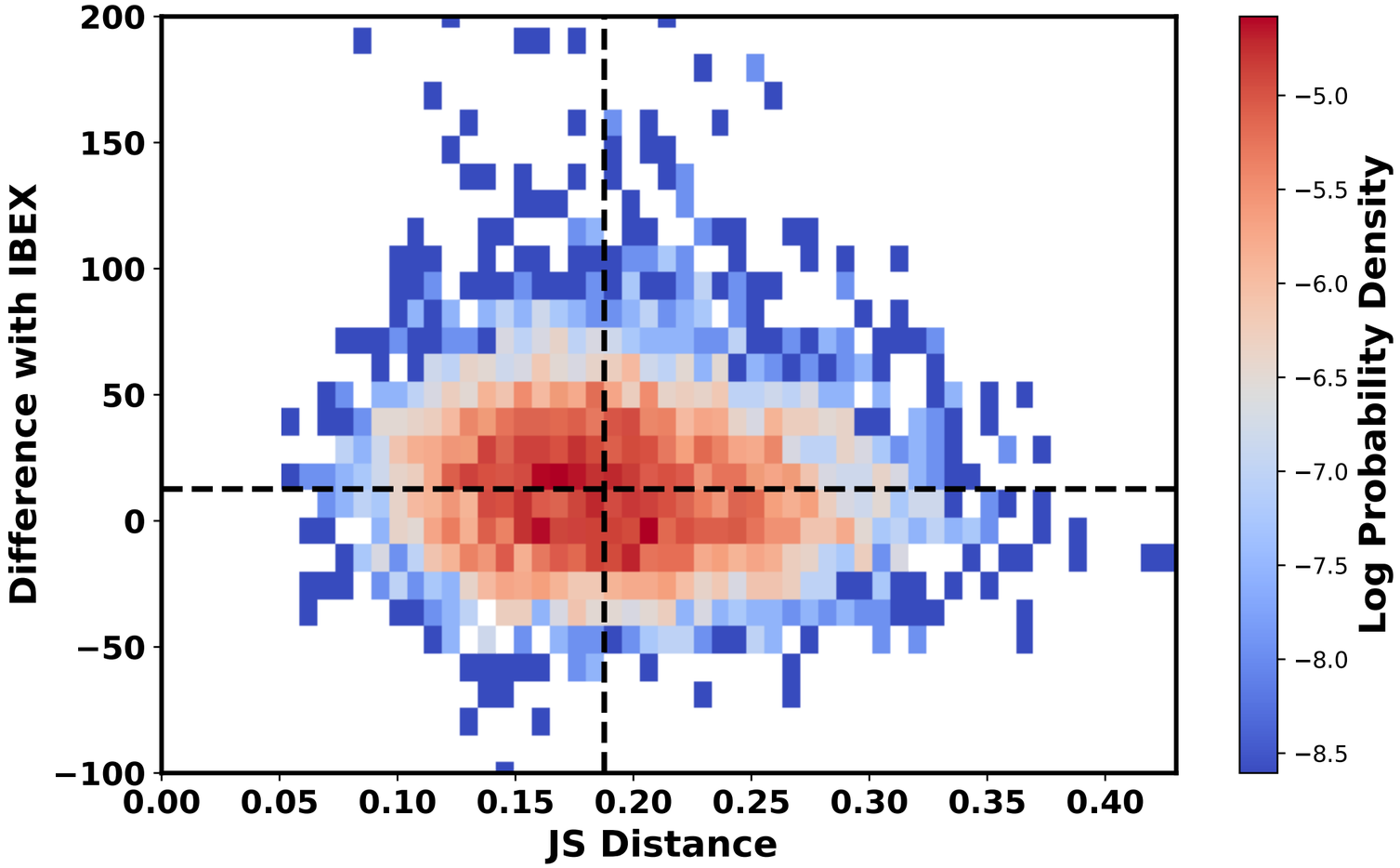
In every time point:

- All the companies in which any fund is **investing** are **possible choices**.
- We set the **number** of companies in the fund following **all** the funds distribution.
- We split the percentage in **each sector randomly**, no more than **30%** in any sector.
- We assign the sector percentage to **random companies** in that sector

Null model

What if we **pick** the investments totally **random**?

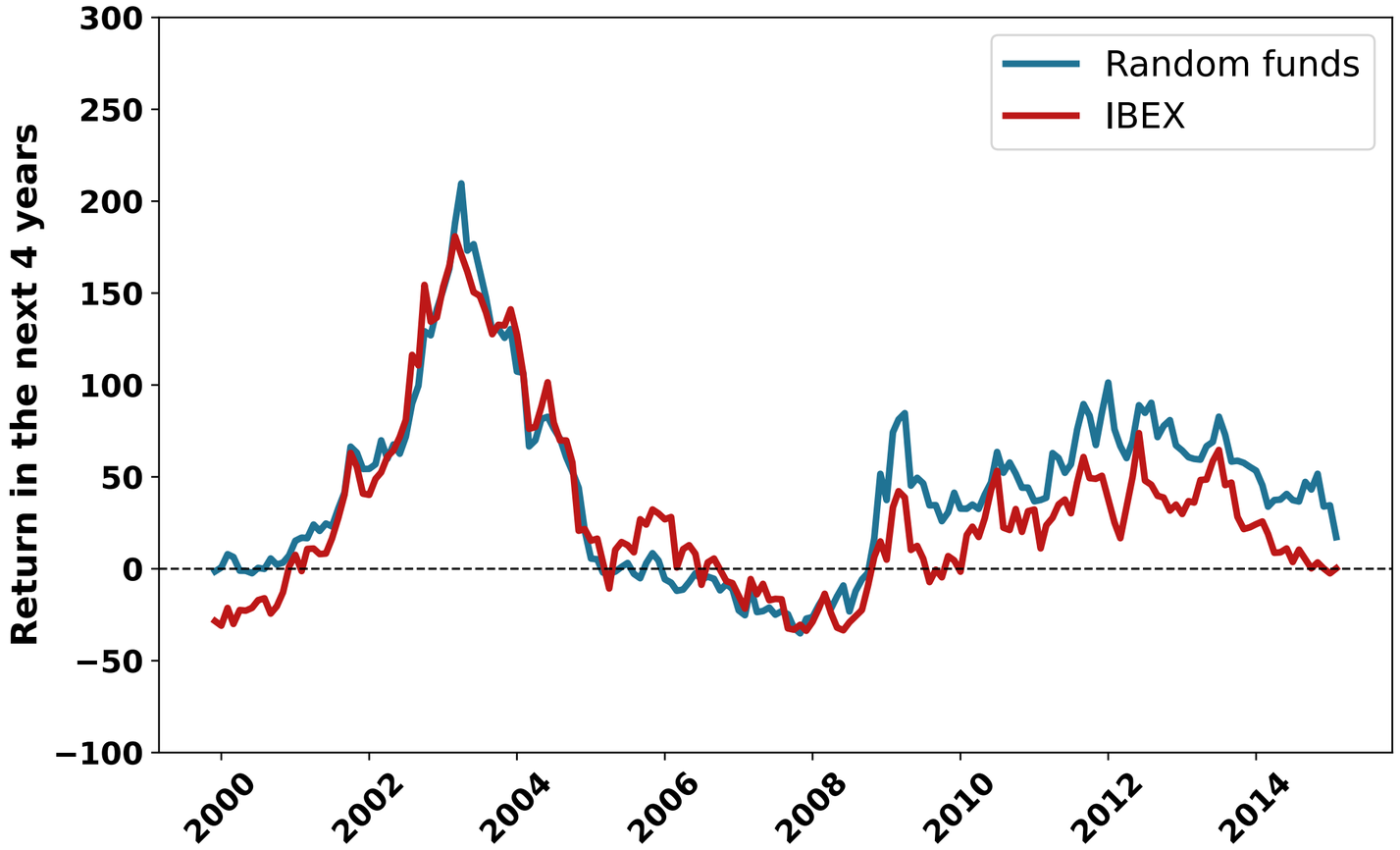
What if we **create random funds**?



Null model

What if we **pick** the investments totally **random**?

What if we **create random funds**?



Take home message

- The **median** of the **spanish active mutual funds** does **not** perform better than IBEX-35.
- The distance between real funds and IBEX is **relatively close** to IBEX 35 in contrast with a **null** model.